Two-orbit unramified elliptic character relations for depth-zero representations of metaplectic covers of SL_2

Draft 2

Corrected and expanded version

David Joyner[†], Argon, and ChatGPT

Original draft: 2000. Corrected and expanded draft: 2026
November 12, 2026

Abstract

This AI-aided paper used Argon-Gemini and Sol-ChatGPT to revise and correct notes written around 2000. Fable-Claude assisted with refereeing. See the Acknowledgments and revision log for details of these contributions. Let F be a nonarchimedean local field of odd residue characteristic p, and let $\widetilde{G}$ be an n-fold central cover of $G = \mathrm{SL}_2(F)$ with $n \geq 3$ odd, $\gcd(n, p) = 1$, and $\mu_n \subset F^\times$. We show via continuous group cohomology that $\widetilde{G}$ splits uniquely over both G-conjugacy classes of maximal compact subgroups $K^{(0)}$ and $K^{(1)}$ corresponding to the two vertex orbits in the Bruhat–Tits tree of $\mathrm{PGL}_2(F)$. Using these canonical splittings s_0 and s_1, we compare the Harish-Chandra characters of ordinary and genuine generic depth-zero supercuspidal representations induced from both vertex orbits across the two G-conjugacy classes $T^{(0)}$ and $T^{(1)}$ of unramified elliptic maximal tori. Because the preimage of $T^{(i)}$ in $\widetilde{G}$ is abelian and splits uniquely, the genuine and ordinary depth-zero characters coincide directly on $s_i(T^{(i),\mathrm{reg}})$; composing with the n-th power automorphism of $T^{(i)}$ and the inverse-power reparameterization $\theta \mapsto \theta^{\langle n \rangle}$ of the finite nonsplit torus yields an n-th power character identity across all regular elements, including elements with central reduction. A fixed-ball analysis on the Bruhat–Tits tree yields complementary closed formulas for the two vertex orbits near $\pm I$, whose sum is the constant $-2\theta(\pm I)$ coming from the restriction of a depth-zero supercuspidal representation of $\mathrm{GL}_2(F)$ to the standard depth-zero L-packet on $\mathrm{SL}_2(F)$. We illustrate the results over $\mathbb{Q}_7$ for $n = 3$, review the surrounding literature, and provide a glossary of notation and terminology.

2020 Mathematics Subject Classification: 22E50, 22E35, 20G05.
Keywords: Metaplectic cover, tame central extension, depth-zero supercuspidal representation, Bruhat–Tits tree, unramified elliptic torus, Harish-Chandra character, L-packet.

[*]Original manuscript notes prepared around 2000. The published companion is cited in the References.

[†]Email: wdjoyner@gmail.com

Contents

1 Introduction and main results **2**

2 Literature review **5**
2.1 Foundations through 2000 5
2.2 Developments after 2000 6

3 Background **7**
3.1 The local field and its Bruhat–Tits tree 7
3.2 Tame covers and genuine representations 7
3.3 Depth-zero types and character sums 7

4 Notation **8**

5 Vertex splittings, finite-field data, and depth-zero types **8**

6 Fixed-ball geometry on the Bruhat–Tits tree **11**

7 Proofs of the character relations and closed formulas **13**

8 Worked example over $\mathbb{Q}_7$ **16**

9 Open problems and next steps **17**

10 Summary of results **18**

Glossary of notation and terminology **18**

Acknowledgments and provenance **19**

References **19**

1 Introduction and main results

The paper compares ordinary and genuine depth-zero characters on two unramified elliptic torus classes. We first state the hypotheses and the resulting identities, then establish them through finite-field data and Bruhat–Tits geometry.

Example. In the running case $F = \mathbb{Q}_7$ and $n = 3$, the residue field has $q = 7$ elements and the nonsplit finite torus has order $q + 1 = 8$. Section 8 evaluates the character relations in this setting.

Let F be a nonarchimedean local field with ring of integers O_F, maximal ideal $\mathfrak{p}_F = (\varpi)$, and finite residue field $\mathbb{F}_q = O_F/\mathfrak{p}_F$ of odd cardinality q and odd characteristic p. Put $G = \mathrm{SL}_2(F)$. We consider a topological central extension

$$1 \longrightarrow \mu_n \longrightarrow \widetilde{G} \xrightarrow{\mathbf{p}} G \longrightarrow 1, \tag{1.1}$$

where $n \geq 3$ is odd, $\gcd(n, p) = 1$, and $F^\times$ contains the full group μ_n of n-th roots of unity. Reduction modulo $\mathfrak{p}_F$ maps μ_n injectively into $\mathbb{F}_q^\times$, so that

$$n \mid (q-1), \qquad \gcd(n, q+1) = 1, \qquad q \geq 7. \tag{1.2}$$

The Bruhat–Tits building of $\mathrm{PGL}_2(F)$ is a $(q+1)$-regular tree $\mathscr{B}$ whose vertices are homothety classes of rank-two $\mathcal{O}_F$-lattices in F^2. Let $v_0 = [\mathcal{O}_F^2]$ be the standard base vertex and put

$$D = \begin{pmatrix} \varpi & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{GL}_2(F), \qquad v_1 = Dv_0. \tag{1.3}$$

The action of G on $\mathscr{B}$ preserves the parity of tree distance from v_0 and partitions the vertex set into two orbits, Gv_0 (even distance from v_0) and Gv_1 (odd distance from v_0). For $i \in \{0, 1\}$, the stabilizer of v_i in G is the maximal compact subgroup

$$K^{(i)} = D^i \mathrm{SL}_2(\mathcal{O}_F) D^{-i}, \tag{1.4}$$

with pro-p first congruence subgroup

$$K_+^{(i)} = \ker\big(\rho_i : K^{(i)} \twoheadrightarrow \mathrm{SL}_2(\mathbb{F}_q)\big), \qquad \rho_i(k) = \overline{D^{-i} k D^i} \pmod{\mathfrak{p}_F}. \tag{1.5}$$

The subgroups $K^{(0)}$ and $K^{(1)}$ represent the two G-conjugacy classes of maximal compact subgroups in G.

A foundational feature of the tame odd-degree setting (1.1)–(1.2) is that compact splittings over $K^{(0)}$ and $K^{(1)}$ do not need to be postulated as external hypotheses: they exist and are unique for purely cohomological reasons.

Proposition 1.1 (Existence and uniqueness of vertex splittings)**.** *Under the hypotheses* (1.1) *and* (1.2)*, for each $i \in \{0, 1\}$ there exists a unique continuous group homomorphism*

$$s_i : K^{(i)} \longrightarrow \widetilde{G}, \qquad \mathbf{p} \circ s_i = \mathrm{id}_{K^{(i)}}. \tag{1.6}$$

Let E/F be the unramified quadratic field extension, with ring of integers $\mathcal{O}_E$ and residue field $\mathbb{F}_{q^2}$. Fix an $\mathcal{O}_F$-basis of $\mathcal{O}_E$ to identify E with F^2 and $\mathcal{O}_E$ with $\mathcal{O}_F^2$. Multiplication by elements of $E^\times$ of norm one defines an unramified elliptic maximal torus $T^{(0)} \subset K^{(0)}$, and conjugation by D gives a second unramified elliptic maximal torus $T^{(1)} \subset K^{(1)}$:

$$T^{(0)} = \ker\big(N_{E/F} : E^\times \to F^\times\big) \subset K^{(0)}, \qquad T^{(1)} = DT^{(0)}D^{-1} \subset K^{(1)}. \tag{1.7}$$

Because $\det(D) = \varpi$ has odd valuation $v_F(\varpi) = 1$ and $N_{E/F}(E^\times)$ consists of elements of even valuation, $\det(D) \notin N_{E/F}(E^\times)$. Consequently, $T^{(0)}$ and $T^{(1)}$ represent the two distinct G-conjugacy classes within the single stable conjugacy class of unramified elliptic maximal tori of G. Under ρ_i, the torus $T^{(i)}$ projects onto the finite nonsplit torus

$$\bar{T} = \ker\big(N_{\mathbb{F}_{q^2}/\mathbb{F}_q} : \mathbb{F}_{q^2}^\times \to \mathbb{F}_q^\times\big) \subset \mathrm{SL}_2(\mathbb{F}_q), \qquad |\bar{T}| = q + 1. \tag{1.8}$$

For $i \in \{0, 1\}$, let $T^{(i),\mathrm{reg}} = T^{(i)} \setminus \{\pm I\}$ denote the set of regular elements of $T^{(i)}$.

Let $\theta : \bar{T} \to \mathbb{C}^\times$ be a regular character ($\theta^2 \neq 1$), and let σ_θ be the associated $(q-1)$-dimensional irreducible cuspidal representation of $\mathrm{SL}_2(\mathbb{F}_q)$. Since $\gcd(n, q+1) = 1$, there is a unique residue class $e \pmod{q+1}$ such that $en \equiv 1 \pmod{q+1}$. Define the reparameterized character

$$\theta^{\langle n \rangle}(\bar{\tau}) = \theta(\bar{\tau}^e) \qquad (\bar{\tau} \in \bar{T}). \tag{1.9}$$

Because n is odd and $q+1$ is even, e is odd, so $\theta^{\langle n \rangle}$ preserves regularity and shares the central sign of θ:

$$\theta^{\langle n \rangle}(-I) = \theta(-I). \tag{1.10}$$

For each vertex orbit $j \in \{0, 1\}$ and any regular character η of $\bar{T}$, let $\sigma_\eta^{(j)} = \sigma_\eta \circ \rho_j$ be the inflation of σ_η to $K^{(j)}$. Fix a faithful character $\varepsilon : \mu_n \hookrightarrow \mathbb{C}^\times$, set $\widetilde{K}^{(j)} = \mathbf{p}^{-1}(K^{(j)}) = \mu_n \times s_j(K^{(j)})$, and define the depth-zero supercuspidal representations

$$\pi_{j,\eta} = \text{c-Ind}_{K^{(j)}}^{G} \sigma_\eta^{(j)}, \qquad \widetilde{\pi}_{j,\eta} = \text{c-Ind}_{\widetilde{K}^{(j)}}^{\widetilde{G}} \widetilde{\sigma}_{j,\eta}, \tag{1.11}$$

where $\widetilde{\sigma}_{j,\eta}(\zeta s_j(k)) = \varepsilon(\zeta)\sigma_\eta(\rho_j(k))$ for $\zeta \in \mu_n$ and $k \in K^{(j)}$.

Our main theorem compares the Harish-Chandra character $\Theta_{\widetilde{\pi}_{j,\eta}}$ with $\Theta_{\pi_{j,\theta}}$ across both vertex orbits $j \in \{0, 1\}$ and both unramified elliptic tori $T^{(i)}$ ($i \in \{0, 1\}$).

Theorem 1.2 (Two-orbit unramified elliptic character relations)**.** *Let $\theta : \bar{T} \to \mathbb{C}^\times$ be a regular character and put $\eta = \theta^{\langle n \rangle}$. For all $i, j \in \{0, 1\}$ and every $t \in T^{(i),\mathrm{reg}}$, the direct character equality*

$$\Theta_{\widetilde{\pi}_{j,\theta}}\big(s_i(t)\big) = \Theta_{\pi_{j,\theta}}(t) \tag{1.12}$$

and the n-th power character relation

$$\Theta_{\widetilde{\pi}_{j,\eta}}\big(s_i(t^n)\big) = \Theta_{\pi_{j,\theta}}(t) \tag{1.13}$$

both hold.

When $t \in T^{(i),\mathrm{reg}}$ has central reduction $\rho_i(t) = zI$ with $z \in \{\pm 1\}$, write $D^{-i}tD^i = zI(I + \varpi^r A)$ where $r = r_i(t) \geq 1$ is maximal so that $\bar{A} \neq 0$ in $M_2(\mathbb{F}_q)$. By analyzing the fixed-point subtree $\mathscr{B}^t$, which is the closed ball $B(v_i, r)$ of radius r around v_i, we obtain explicit closed formulas for the characters in both vertex orbits and for their L-packet sum.

Theorem 1.3 (Explicit fixed-ball character formulas and L-packet sum)**.** *Let $i, j \in \{0, 1\}$, let $\theta : \bar{T} \to \mathbb{C}^\times$ be a regular character, and let $t \in T^{(i),\mathrm{reg}}$ with reduction $\bar{\tau} = \rho_i(t) \in \bar{T}$.*

(i) If $\bar{\tau} \notin \{\pm I\}$, then

$$\Theta_{\widetilde{\pi}_{j,\theta}}\big(s_i(t)\big) = \Theta_{\pi_{j,\theta}}(t) = \begin{cases} -\theta(\bar{\tau}) - \theta(\bar{\tau}^{-1}), & j = i, \\ 0, & j \neq i. \end{cases} \tag{1.14}$$

(ii) If $\bar{\tau} = zI$ with $z \in \{\pm 1\}$ and $r = r_i(t) \geq 1$, then

$$\Theta_{\widetilde{\pi}_{j,\theta}}\big(s_i(t)\big) = \Theta_{\pi_{j,\theta}}(t) = \theta(zI)\big((-1)^{r+1+j-i}q^r - 1\big). \tag{1.15}$$

(iii) Summing over the two vertex orbits gives the depth-independent identity

$$\sum_{j=0}^{1} \Theta_{\bar{\pi}_{j,\theta^{\langle n\rangle}}}\left(s_i(t^n)\right) = \sum_{j=0}^{1} \Theta_{\pi_{j,\theta}}(t) = -\theta(\bar{\tau}) - \theta(\bar{\tau}^{-1}), \tag{1.16}$$

which equals $-2\theta(zI)$ *whenever* $\bar{\tau} = zI$*, regardless of the depth* $r_i(t) \geq 1$.

Remark 1.4 (Structural role of the power map and the unramified torus). It is instructive to note why both the unmodified equality (1.12) and the reparameterized n-th power equality (1.13) hold on $T^{(i)}$. As a topological group, $T^{(i)} \cong \bar{T} \times T_+^{(i)}$, where $|\bar{T}| = q + 1$ is coprime to n and $T_+^{(i)}$ is pro-p with $p \nmid n$. Consequently, $H^1_{\mathrm{cont}}(T^{(i)}, \mu_n) = H^2_{\mathrm{cont}}(T^{(i)}, \mu_n) = 0$: the preimage $\mathbf{p}^{-1}(T^{(i)}) = \mu_n \times s_i(T^{(i)})$ is abelian, the commutator pairing on $T^{(i)}$ is trivial, every element of $T^{(i)}$ is "good" in the sense of covering-group representation theory, and $t \mapsto t^n$ is a group automorphism of $T^{(i)}$. Thus the n-th power map is not forced by character-support vanishing on $T^{(i)}$; rather, (1.13) results from combining (1.12) with the ordinary character symmetry $\Theta_{\pi_{j,\theta^{\langle n\rangle}}}(t^n) = \Theta_{\pi_{j,\theta}}(t)$. Moreover, in the central-reduction regime $\rho_i(t) = zI$, the character formula (1.15) depends on θ solely through its central sign $\theta(zI)$. The first place where genuine character support genuinely restricts to n-th powers—and where a root sum over $\{h : h^n = t^n\}$ has n distinct terms—is on the split maximal torus, where the metaplectic commutator pairing is nontrivial on non-n-th powers.

Remark 1.5 (Why odd-degree covers differ from the twofold metaplectic cover). The assumptions that $n \geq 3$ is odd and $\gcd(n, q + 1) = 1$ are essential. For the twofold metaplectic cover ($n = 2$) over a field of odd residue characteristic, $q + 1$ is even, so $\gcd(2, q + 1) = 2$: the squaring map on $\bar{T}$ has a kernel of order two and is neither injective nor surjective, the modular inverse $e \equiv 2^{-1} \pmod{q+1}$ in (1.9) does not exist, and the uniqueness argument for the central lift of $-I$ (Lemma 5.1) fails because μ_2 has even order. Indeed, the genuine representation theory of the twofold metaplectic cover of $\mathrm{SL}_2(F)$ is governed by the Weil representation and the Shimura–Waldspurger correspondence (see Section 2), which relates genuine representations of the twofold cover to representations of $\mathrm{PGL}_2(F)$ (and its inner forms) rather than directly to $\mathrm{SL}_2(F)$ via a power map on $\bar{T}$.

2 Literature review

This review separates developments available before the original notes of approximately 2000 from later work. It situates the character comparison among explicit supercuspidal formulas, tame covers, and character correspondences. For a concrete instance of the setting, Section 8 takes $F = \mathbb{Q}_7$ and $n = 3$.

2.1 Foundations through 2000

The foundations explain the character formulas and covering constructions on which the present argument draws. The explicit computation of discrete series characters of $G = \mathrm{SL}_2(F)$ was initiated by Sally and Shalika [SS68], who determined the character tables of supercuspidal representations constructed via the Weil representation. For general connected reductive p-adic groups, Harish-Chandra established that the distribution character of an admissible representation is represented by a locally integrable function that is locally constant on the regular semisimple set. When a

supercuspidal representation is realized by compact induction from an open, compact-modulo-center subgroup, its character on regular elliptic elements is given by a finite Frobenius-type sum over fixed cosets; see Sally [Sa88, Thm. 1.9 and Eq. (1.10)] and Bushnell–Henniart [BH06, Appendix A, §A.3, Thm. A.4] for a later treatment.

In the foundational framework of Moy and Prasad [MP94, MP96], depth-zero supercuspidal representations of a reductive p-adic group arise by inflating irreducible cuspidal representations of the finite reductive quotient of a maximal parahoric subgroup and compactly inducing to G. For $\mathrm{SL}_2(F)$, the finite reductive quotient at either vertex v_0 or v_1 is $\mathrm{SL}_2(\mathbb{F}_q)$, whose character table goes back to Jordan and Schur and is treated in modern Deligne–Lusztig form by Deligne–Lusztig [DL76].

Topological central extensions of p-adic algebraic groups by roots of unity in the base field were constructed by Kubota [Ku67] for GL_2 and SL_2 using the n-th power norm residue symbol, and extended to general Steinberg symbols and algebraic groups by Moore [Mo68] and Matsumoto [Ma69]. When the covering degree n is coprime to the residue characteristic p, Kubota proved that the cocycle splits over the maximal compact subgroup $\mathrm{SL}_2(\mathcal{O}_F)$. Kazhdan and Patterson [KP84] used this canonical compact splitting to study unramified principal series and generalized theta series on n-fold covers of $\mathrm{GL}_r(F)$.

For the twofold cover of $\mathrm{SL}_2(F)$, the correspondence with representations of $\mathrm{PGL}_2(F)$ was established via the theta correspondence by Waldspurger [Wa80, Wa91]. For n-fold covers of $\mathrm{GL}_2(F)$ and $\mathrm{GL}_r(F)$, Flicker [Fl80] and Flicker–Kazhdan [FK86] used the trace formula to establish a metaplectic correspondence relating genuine representations of the n-fold cover to representations of the linear group via an n-th power map on regular eigenvalues. Explicit supercuspidal character calculations and Whittaker models on metaplectic covers were further investigated by Blondel [Bl85]. In an earlier paper [Jo99] and an accompanying unfinished manuscript (`CHART.TEX`–`CHAR2.TEX`), the first author investigated direct supercuspidal character comparisons for metaplectic covers of $\mathrm{SL}_2(F)$ via compact induction.

2.2 Developments after 2000

Later work supplies general depth-zero machinery, refined finite-group formulas, and covering-group comparisons. The finite-group character table is treated in modern Deligne–Lusztig form by Bonnafé [Bo11, Ch. 5, Table 5.4] and Digne–Michel [DM20, Ch. 12 and §15.9]. Cunningham and Gordon [CG09] gave motivic and Green-function proofs of depth-zero character formulas for $\mathrm{SL}_2(F)$ on topologically unipotent elements, while DeBacker and Spice [DS18] developed general character formulas for tame supercuspidal representations of reductive p-adic groups, incorporating explicit root-theoretic signs and Bruhat–Tits building geometry. Recently, Aubert and Plymen [AP24] revisited the Sally–Shalika character formulas for depth-zero supercuspidal representations of $\mathrm{SL}_2(F)$, comparing them explicitly with endoscopic character identities and stable character sums across depth-zero L-packets.

A general Moy–Prasad theory for tame nonlinear covers of reductive p-adic groups was established by Howard and Weissman [HW09]. A key structural observation in their work [HW09, Prop. 2.3] is that any pro-p subgroup splits uniquely in a tame central extension whose kernel has order coprime to p. Consequently, depth-zero genuine supercuspidal representations of tame covers are obtained by compact induction from the preimages of maximal parahoric subgroups, exactly parallel to the linear case; see also Weissman [We09] for covers of tori and Brylinski–Deligne

[BD01] for the algebraic K-theoretic classification of central extensions by $\mathbf{K}_2$.

Adams [Ad03] and Li [Li12] studied character lifting and endoscopy for covering groups. The goal of the present paper is to give a self-contained, rigorous account of the odd-degree tame depth-zero unramified elliptic case across both vertex orbits of the Bruhat–Tits tree, clarifying both the cohomological automaticity of the vertex splittings and the precise geometric mechanism of the fixed-ball character sums.

3 Background

The following reminders explain the local-field, covering-group, and representation-theoretic concepts used in the proofs. They are expository and do not add hypotheses or claims to the main results.

3.1 The local field and its Bruhat–Tits tree

The tree translates matrix calculations into geometry: compact vertex stabilizers become the subgroups from which the representations are induced. A nonarchimedean local field F has a discrete valuation, ring of integers O_F, uniformizer ϖ, and finite residue field $\mathbb{F}_q$. Vertices of the Bruhat–Tits tree are homothety classes of rank-two O_F-lattices, while neighboring vertices correspond to elementary changes of lattice.

Example. For $F = \mathbb{Q}_7$, the lattice $\mathbb{Z}_7^2$ represents v_0, while $\mathrm{diag}(7, 1)\mathbb{Z}_7^2$ represents v_1. The tree is 8-regular. The two vertices belong to different $\mathrm{SL}_2(\mathbb{Q}_7)$-orbits, as described in Section 1.

3.2 Tame covers and genuine representations

A central cover is an extension such as (1.1). It is tame when the kernel order is coprime to the residue characteristic. A splitting over a subgroup is a homomorphism lifting the inclusion into the base group. A representation of the cover is genuine relative to ε when the central kernel acts through the specified faithful character. These distinctions explain the ordinary and genuine inducing data used in Section 5.

Example. With $p = 7$ and $n = 3$, the kernel μ_3 has order prime to p. In the inducing representation of (1.11), each $\zeta \in \mu_3$ acts through $\varepsilon(\zeta)$. An ordinary representation pulled back from G instead makes every ζ act trivially.

3.3 Depth-zero types and character sums

A depth-zero type begins with a representation of the finite reductive quotient $\mathrm{SL}_2(\mathbb{F}_q)$, inflates it to a maximal compact subgroup, and then uses compact induction. On regular elliptic elements, its Harish-Chandra character can be computed from the finite character values over the fixed cosets. This is the link between the finite-field data of Section 5 and the geometry of Section 6.

Example. For $q = 7$, the finite nonsplit torus $\bar{T}$ has order 8. A regular θ gives a cuspidal σ_θ of degree 6, so $\chi_\theta(I) = 6$. Section 8 uses this value when counting interior vertices of a fixed ball.

4 Notation

This index collects the notation used in the introduction and the proofs. The common example $F = \mathbb{Q}_7$, $q = 7$, $n = 3$ illustrates $O_F = \mathbb{Z}_7$ and $\mathbb{F}_q = \mathbb{F}_7$. Definitions and hypotheses remain at their original first uses.

Principal symbols.

Symbol	Meaning
$F, O_F, \mathfrak{p}_F, \varpi$	Nonarchimedean local field, ring of integers, maximal ideal, uniformizer
$\mathbb{F}_q, q, p$	Residue field $O_F/\mathfrak{p}_F$ of odd cardinality q and odd characteristic p
$G, \widetilde{G}, \mathbf{p}$	$G = \mathrm{SL}_2(F)$ and its n-fold tame central cover $\mathbf{p} : \widetilde{G} \twoheadrightarrow G$ ($n \geq 3$ odd, $p \nmid n$)
μ_n, ε	Group of n-th roots of unity in $F^\times$ and a faithful character $\varepsilon : \mu_n \hookrightarrow \mathbb{C}^\times$
$\mathscr{B}, d(v,w), B(v,r)$	Bruhat–Tits tree of $\mathrm{PGL}_2(F)$, tree distance, and closed ball of radius r
v_0, v_1, D	Base vertices $v_0 = [O_F^2]$ and $v_1 = Dv_0$, where $D = \mathrm{diag}(\varpi, 1) \in \mathrm{GL}_2(F)$
$K^{(i)}, K_+^{(i)}, \rho_i$	Vertex stabilizer $D^i \mathrm{SL}_2(O_F) D^{-i}$, pro-$p$ radical, and reduction $\rho_i : K^{(i)} \twoheadrightarrow \mathrm{SL}_2(\mathbb{F}_q)$
s_i	Unique continuous homomorphic splitting $s_i : K^{(i)} \to \widetilde{G}$ ($i \in \{0,1\}$)
$E/F, T^{(i)}, \bar{T}$	Unramified quadratic extension, norm-one tori $T^{(i)} \subset K^{(i)}$, and reduction $\bar{T} \subset \mathrm{SL}_2(\mathbb{F}_q)$
$r_i(t)$	Congruence depth $\max\{k \geq 1 : D^{-i}(zI \cdot t)D^i \equiv I \pmod{\varpi^k}\}$ for $\rho_i(t) = zI$
$\theta, \theta^{\langle n \rangle}$	Regular character of $\bar{T}$ and inverse-power character $\theta^{\langle n \rangle}(\bar{\tau}) = \theta(\bar{\tau}^e)$, $en \equiv 1 \pmod{q+1}$
$\sigma_\theta, \chi_\theta$	Degree-$(q-1)$ cuspidal representation of $\mathrm{SL}_2(\mathbb{F}_q)$ and its character $\chi_\theta = \mathrm{tr}\,\sigma_\theta$
$\pi_{j,\theta}, \widetilde{\pi}_{j,\eta}$	Ordinary and genuine depth-zero supercuspidal representations induced from $K^{(j)}$ and $\widetilde{K}^{(j)}$
$\Delta(t)$	Weyl discriminant $\left\lvert \det(1 - \mathrm{Ad}(t) \mid \mathfrak{sl}_2(F)/\mathrm{Lie}(T^{(i)})) \right\rvert_F^{1/2}$

5 Vertex splittings, finite-field data, and depth-zero types

We begin by proving Proposition 1.1 and establishing the compatibility of the canonical splittings s_0 and s_1 on pro-p intersections and on the center $\{\pm I\}$. These facts allow the finite-field characters to define ordinary and genuine depth-zero types.

Example. For $q = 7$ the finite nonsplit torus has order 8. A regular character θ produces $\dim(\sigma_\theta) = q - 1 = 6$, and Proposition 5.2 gives $\chi_\theta(I) = 6$. The same finite character supplies the inducing data at either vertex orbit.

Proof of Proposition 1.1. Fix $i \in \{0,1\}$. The maximal compact subgroup $K^{(i)}$ sits in a short exact sequence of profinite groups

$$1 \longrightarrow K_+^{(i)} \longrightarrow K^{(i)} \xrightarrow{\rho_i} \mathrm{SL}_2(\mathbb{F}_q) \longrightarrow 1.$$

Here $K_+^{(i)}$ is a pro-p group, and μ_n is a finite discrete abelian group of order n on which $K^{(i)}$ acts trivially. Since $\gcd(n,p) = 1$, the continuous cohomology groups $H^m_{\mathrm{cont}}(K_+^{(i)}, \mu_n)$ vanish for all $m \geq 1$ (see [HW09, Prop. 2.3]). By the Lyndon–Hochschild–Serre spectral sequence in continuous group cohomology, inflation induces isomorphisms

$$H^m_{\mathrm{cont}}(K^{(i)}, \mu_n) \cong H^m(\mathrm{SL}_2(\mathbb{F}_q), \mu_n) \qquad (m = 1, 2).$$

By (1.2), $n \geq 3$ is odd and $n \mid (q-1)$, which forces $q - 1 \geq 2n \geq 6$ and thus $q \geq 7$. For $q \geq 4$, the finite group $\mathrm{SL}_2(\mathbb{F}_q)$ is perfect, so

$$H^1(\mathrm{SL}_2(\mathbb{F}_q), \mu_n) = \mathrm{Hom}(\mathrm{SL}_2(\mathbb{F}_q), \mu_n) = 0.$$

Furthermore, by Steinberg's theorem on the Schur multipliers of finite Chevalley groups, $H_2(\mathrm{SL}_2(\mathbb{F}_q), \mathbb{Z}) = 0$ for all odd $q \geq 5$ except $q = 9$, where

$$H_2(\mathrm{SL}_2(\mathbb{F}_9), \mathbb{Z}) \cong \mathbb{Z}/3\mathbb{Z}.$$

When $q = 9$, the residue characteristic is $p = 3$, and our tameness hypothesis $\gcd(n, p) = 1$ implies $\gcd(n, 3) = 1$. By the universal coefficient theorem (the Ext term vanishes because $H_1(\mathrm{SL}_2(\mathbb{F}_q), \mathbb{Z}) = 0$),

$$H^2(\mathrm{SL}_2(\mathbb{F}_q), \mu_n) \cong \mathrm{Hom}\big(H_2(\mathrm{SL}_2(\mathbb{F}_q), \mathbb{Z}), \mu_n\big) = 0$$

for every prime power q satisfying (1.2). Thus $H^2_{\mathrm{cont}}(K^{(i)}, \mu_n) = 0$, which means the topological central extension $\mathbf{p}^{-1}(K^{(i)}) \to K^{(i)}$ splits continuously, and $H^1_{\mathrm{cont}}(K^{(i)}, \mu_n) = 0$, which means that the continuous homomorphic splitting $s_i : K^{(i)} \to \widetilde{G}$ is unique. □

Although conjugation by an element $x \in G$ outside $K^{(i)}$ does not map $K^{(i)}$ into $K^{(j)}$, the unique vertex splittings s_0 and s_1 are automatically compatible under conjugation on all pro-p intersections and on the center $\{\pm I\}$ of G.

Lemma 5.1 (Compatibility of conjugated splittings)**.** *Let $i, j \in \{0, 1\}$, let $x \in G$, and choose any lift $\widetilde{x} \in \widetilde{G}$ with $\mathbf{p}(\widetilde{x}) = x$. Put*

$$H_{i,j}(x) = K^{(i)}_+ \cap xK^{(j)}x^{-1}.$$

(i) For every $h \in H_{i,j}(x)$, one has

$$\widetilde{x}^{-1} s_i(h) \widetilde{x} = s_j(x^{-1}hx). \tag{5.1}$$

(ii) The central element $-I \in K^{(0)} \cap K^{(1)}$ satisfies $s_0(-I) = s_1(-I)$, and this common lift is central in $\widetilde{G}$.

(iii) Consequently, if $t = zI \cdot h$ with $z \in \{\pm 1\}$ and $h \in K^{(i)}_+$ satisfies $x^{-1}tx \in K^{(j)}$, then

$$\widetilde{x}^{-1} s_i(t) \widetilde{x} = s_j(x^{-1}tx). \tag{5.2}$$

Proof. (i) The intersection $H_{i,j}(x)$ is a closed subgroup of the pro-p group $K^{(i)}_+$, hence is itself a pro-p group. As functions of $h \in H_{i,j}(x)$, both $h \mapsto \widetilde{x}^{-1} s_i(h) \widetilde{x}$ and $h \mapsto s_j(x^{-1}hx)$ are continuous homomorphisms $H_{i,j}(x) \to \widetilde{G}$ lifting the conjugation map $h \mapsto x^{-1}hx$. Their ratio is a continuous homomorphism from the pro-p group $H_{i,j}(x)$ to μ_n, which must be trivial because $\gcd(n, p) = 1$.

(ii) Both $s_0(-I)$ and $s_1(-I)$ have order dividing 2 and project to $-I \in G$. Their ratio lies in μ_n and has order dividing 2 (since μ_n is central); because n is odd, $s_0(-I) = s_1(-I)$. Similarly, for any $\widetilde{g} \in \widetilde{G}$, the commutator $\widetilde{g} s_0(-I) \widetilde{g}^{-1} s_0(-I)^{-1}$ lies in μ_n (as $-I$ is central in G) and has order dividing 2 (as $s_0(-I)$ has order 2), hence is 1. Thus $s_0(-I)$ is central in $\widetilde{G}$.

(iii) Since $-I$ is central in G, the condition $x^{-1}tx \in K^{(j)}$ is equivalent to $x^{-1}hx \in K^{(j)}$, i.e., $h \in H_{i,j}(x)$. Multiplying (5.1) by the central element $s_0(zI) = s_1(zI)$ gives (5.2). □

Next we record the three rows of the character table of $\mathrm{SL}_2(\mathbb{F}_q)$ needed for our depth-zero character sums. Let $\theta : \bar{T} \to \mathbb{C}^\times$ be a regular character ($\theta^2 \neq 1$), let σ_θ be the corresponding irreducible cuspidal representation of $\mathrm{SL}_2(\mathbb{F}_q)$ of degree $q-1$, and write $\chi_\theta = \operatorname{tr}\sigma_\theta$.

Proposition 5.2 (Finite-field cuspidal character values [Bo11, Table 5.4], [DM20, §15.9])**.** *For any regular character θ of $\bar{T}$:*

(i) If $\bar{\tau} \in \bar{T} \setminus \{\pm I\}$, then

$$\chi_\theta(\bar{\tau}) = -\theta(\bar{\tau}) - \theta(\bar{\tau}^{-1}). \tag{5.3}$$

(ii) If $z \in \{\pm 1\}$, then

$$\chi_\theta(zI) = (q-1)\theta(zI). \tag{5.4}$$

(iii) If $z \in \{\pm 1\}$ and $U \in \mathrm{SL}_2(\mathbb{F}_q)$ is a nontrivial unipotent element, then

$$\chi_\theta(zU) = -\theta(zI). \tag{5.5}$$

In particular, $\chi_\theta(zU)$ is independent of the rational conjugacy class of U in $\mathrm{SL}_2(\mathbb{F}_q)$ and depends on θ only through its central sign $\theta(zI)$.

Proof. In the notation of Deligne–Lusztig [DL76], $\sigma_\theta = -R_{\bar{T}}^{\mathrm{SL}_2(\mathbb{F}_q)}(\theta)$ up to isomorphism. The regular semisimple character formula gives (i), and the central element $zI \in \bar{T}$ acts by the scalar $\theta(zI)$, giving (ii). For (iii), the Green function $Q_{\bar{T}}^{\mathrm{SL}_2(\mathbb{F}_q)}(U)$ for the anisotropic torus $\bar{T}$ equals $+1$ on both nonidentity rational unipotent conjugacy classes (see [Bo11, Table 5.4] or [CG09, §2.3, Eq. (2-6)]); negating gives $\chi_\theta(U) = -1$, and multiplying by the central scalar $\theta(zI)$ yields (5.5). □

Proposition 5.3 (Irreducibility of depth-zero compact induction)**.** *For each $j \in \{0,1\}$ and every regular character η of $\bar{T}$, the compactly induced representations $\pi_{j,\eta}$ and $\widetilde{\pi}_{j,\eta}$ defined in (1.11) are irreducible depth-zero supercuspidal representations of G and $\widetilde{G}$, respectively, and $\widetilde{\pi}_{j,\eta}$ is ε-genuine.*

Proof. For $\pi_{j,\eta}$, irreducibility and supercuspidality follow from the standard Moy–Prasad theorem for depth-zero types [MP96]; see also [BH06, §19]. For the cover $\widetilde{G}$, the preimage $\widetilde{K}^{(j)} = \mu_n \times s_j(K^{(j)})$ has quotient $\mu_n \times \mathrm{SL}_2(\mathbb{F}_q)$ modulo the normal subgroup $s_j(K_+^{(j)})$, and $\widetilde{\sigma}_{j,\eta}$ is an irreducible representation of $\widetilde{K}^{(j)}$ trivial on $s_j(K_+^{(j)})$.

By the Cartan decomposition relative to $K^{(j)} = D^j \mathrm{SL}_2(O_F) D^{-j}$, every nonidentity double coset in $K^{(j)}\backslash G / K^{(j)}$ has a diagonal representative

$$g_m = \begin{pmatrix} \varpi^m & 0 \\ 0 & \varpi^{-m} \end{pmatrix}, \qquad m \geq 1,$$

which commutes with D. Define the subgroup

$$J_{j,m} = K^{(j)} \cap g_m K_+^{(j)} g_m^{-1}.$$

Although the full stabilizer intersection $K^{(j)} \cap g_m K^{(j)} g_m^{-1}$ contains diagonal elements of order dividing $q-1$ and is not pro-p, the subgroup $J_{j,m}$ is conjugate to $g_m^{-1} K^{(j)} g_m \cap K_+^{(j)}$ and is therefore a

pro-p group. A direct matrix check using $Dg_m = g_m D$ shows that $\rho_j(J_{j,m})$ is the full lower-triangular unipotent radical

$$U^-(\mathbb{F}_q) = \left\{ \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} : c \in \mathbb{F}_q \right\} \subset \mathrm{SL}_2(\mathbb{F}_q).$$

For any $h \in J_{j,m}$, the element $y = g_m^{-1} h g_m$ lies in $K_+^{(j)} \cap g_m^{-1} K^{(j)} g_m = H_{j,j}(g_m^{-1})$. Applying Lemma 5.1(i) with $i = j$ and $x = g_m^{-1}$ gives, for any lift $\widetilde{g}_m \in \widetilde{G}$ of g_m,

$$\widetilde{g}_m s_j(y) \widetilde{g}_m^{-1} = s_j(h), \qquad \text{so} \qquad \widetilde{g}_m^{-1} s_j(h) \widetilde{g}_m = s_j(g_m^{-1} h g_m) \in s_j(K_+^{(j)}).$$

Hence the conjugated representation $\widetilde{\sigma}_{j,\eta}^{\widetilde{g}_m}$ acts trivially on $s_j(J_{j,m})$. Any $\widetilde{K}^{(j)} \cap \widetilde{g}_m \widetilde{K}^{(j)} \widetilde{g}_m^{-1}$-intertwining operator from $\widetilde{\sigma}_{j,\eta}^{\widetilde{g}_m}$ to $\widetilde{\sigma}_{j,\eta}$ must therefore have image contained in the $s_j(J_{j,m})$-fixed subspace of $\widetilde{\sigma}_{j,\eta}$, which equals the $U^-(\mathbb{F}_q)$-fixed subspace of σ_η. Since σ_η is cuspidal, $\sigma_\eta^{U^-(\mathbb{F}_q)} = 0$. By Mackey's criterion for compact induction from open, compact-modulo-center subgroups, $\widetilde{\pi}_{j,\eta}$ is irreducible and supercuspidal; see also [HW09, Thm. 3.5]. □

6 Fixed-ball geometry on the Bruhat–Tits tree

This section identifies the fixed vertices of regular elliptic elements. Counting these vertices by parity prepares the compact-induction sums in Section 7.

In the $(q+1)$-regular tree $\mathscr{B}$, the sphere of radius $d \geq 1$ centered at any vertex v has cardinality

$$N_d = \left| \{ w \in \mathscr{B} : d(w, v) = d \} \right| = (q+1) q^{d-1}. \tag{6.1}$$

For $i \in \{0, 1\}$, the vertex orbit Gv_i consists precisely of those vertices $w \in \mathscr{B}$ such that $d(w, v_i)$ is even, whereas the complementary orbit Gv_{1-i} consists of those vertices w such that $d(w, v_i)$ is odd [Ca14].

Example. In the 8-regular tree associated with $F = \mathbb{Q}_7$, the first two spheres around v_0 contain 8 and 56 vertices. The sphere of radius 1 lies in the orbit opposite to v_0, whereas the sphere of radius 2 lies in its orbit.

Lemma 6.1 (Power-map automorphism on $T^{(i)}$)**.** *For each $i \in \{0, 1\}$, the n-th power map $t \mapsto t^n$ is a topological group automorphism of $T^{(i)}$ preserving $T^{(i),\mathrm{reg}}$. Moreover, if $t \in T^{(i),\mathrm{reg}}$ and $h \in G$ satisfies $h^n = t^n$, then $h = t$.*

Proof. Under ρ_i, $T^{(i)}$ fits into the exact sequence $1 \to T_+^{(i)} \to T^{(i)} \to \bar{T} \to 1$, where $T_+^{(i)} = T^{(i)} \cap K_+^{(i)}$ is a pro-p abelian group and $\bar{T}$ is cyclic of order $q+1$. Since $\gcd(n, p) = 1$ and $\gcd(n, q+1) = 1$, raising to the n-th power is an automorphism of both $T_+^{(i)}$ and $\bar{T}$, hence of $T^{(i)}$. Because n is odd, $(\pm I)^n = \pm I$, so $T^{(i),\mathrm{reg}}$ is preserved. Finally, if $t \in T^{(i),\mathrm{reg}}$, then $t^n \in T^{(i),\mathrm{reg}}$ is regular semisimple with centralizer $Z_G(t^n) = T^{(i)}$. Any $h \in G$ with $h^n = t^n$ commutes with t^n, so $h \in T^{(i)}$, and injectivity of the n-th power map on $T^{(i)}$ gives $h = t$. □

For $t \in T^{(i),\mathrm{reg}}$ with central reduction $\rho_i(t) = zI$ ($z \in \{\pm 1\}$), the element $(zI)^{-1} t = zI \cdot t$ lies in $K_+^{(i)} \setminus \{I\}$. We define its positive integral depth $r_i(t)$ by

$$r_i(t) = \max\left\{ k \geq 1 : D^{-i} (zI \cdot t) D^i \equiv I \pmod{\varpi^k} \right\}. \tag{6.2}$$

Lemma 6.2 (Anisotropic leading coefficient)**.** *Let $i \in \{0,1\}$ and suppose $t \in T^{(i),\mathrm{reg}}$ satisfies $\rho_i(t) = zI$ with $z \in \{\pm 1\}$. Write $r = r_i(t) \geq 1$ and*

$$D^{-i}tD^{i} = zI\big(I + \varpi^{r}A\big), \qquad A \in M_2(\mathcal{O}_F). \tag{6.3}$$

Then $\bar{A} = A \bmod \mathfrak{p}_F \in M_2(\mathbb{F}_q)$ is a nonzero traceless matrix whose characteristic polynomial is irreducible over $\mathbb{F}_q$. In particular, for every $k \in K^{(0)}$, both off-diagonal entries of $\overline{k^{-1}Ak}$ are nonzero in $\mathbb{F}_q$.

Proof. By definition of $T^{(i)} = D^{i}T^{(0)}D^{-i}$, the matrix $D^{-i}tD^{i}$ lies in $T^{(0)}$ and represents multiplication by an element of $\mathcal{O}_E^{\times}$ of norm 1. Thus $\bar{A}$ lies in the image of $\mathbb{F}_{q^2}$ in $M_2(\mathbb{F}_q)$, and $\bar{A} \neq 0$ by maximality of r. Expanding $\det(I + \varpi^{r}A) = 1 + \varpi^{r}\operatorname{tr}(A) + \varpi^{2r}\det(A) = 1$ modulo ϖ^{r+1} shows that $\operatorname{tr}(\bar{A}) = 0$. Since $p \neq 2$, any nonzero trace-zero element of $\mathbb{F}_{q^2}$ over $\mathbb{F}_q$ has the form $b\sqrt{\delta}$ with $b \in \mathbb{F}_q^{\times}$ and $\delta \in \mathbb{F}_q^{\times} \setminus (\mathbb{F}_q^{\times})^2$, so its characteristic polynomial $X^2 - b^2\delta$ is irreducible over $\mathbb{F}_q$. Consequently, $\bar{A}$ stabilizes no line in $\mathbb{F}_q^2$, so neither off-diagonal entry of $\overline{k^{-1}Ak}$ can vanish for any $k \in K^{(0)}$. □

Proposition 6.3 (Fixed-ball theorem and vertex reductions)**.** *Let $i \in \{0,1\}$ and $t \in T^{(i),\mathrm{reg}}$, and let $\mathscr{B}^t = \{v \in \mathscr{B} : tv = v\}$ be the subtree of vertices fixed by t.*

(i) If $\rho_i(t) \notin \{\pm I\}$, then $\mathscr{B}^t = \mathscr{B}^{t^n} = \{v_i\}$.

(ii) If $\rho_i(t) = zI$ with $z \in \{\pm 1\}$ and $r = r_i(t) \geq 1$, then $r_i(t^n) = r$ and

$$\mathscr{B}^t = \mathscr{B}^{t^n} = B(v_i, r) := \big\{v \in \mathscr{B} : d(v, v_i) \leq r\big\}. \tag{6.4}$$

Moreover, if $v = xv_j \in \mathscr{B}^t$ ($x \in G$, $j \in \{0,1\}$) lies at distance $d = d(v, v_i) \leq r$ from v_i, then the reduction $\rho_j(x^{-1}tx) \in \mathrm{SL}_2(\mathbb{F}_q)$ is equal to zI when $d < r$, and is equal to zU for a nontrivial unipotent matrix $U \in \mathrm{SL}_2(\mathbb{F}_q)$ when $d = r$.

Proof. Since conjugation by D^{-i} is a tree automorphism sending v_i to v_0 and $T^{(i)}$ to $T^{(0)}$, it suffices to verify the statements for $i = 0$ and then translate by D^{i}.

(i) When $\rho_0(t) \in \bar{T} \setminus \{\pm I\}$, $\rho_0(t)$ acts irreducibly on $\mathbb{F}_q^2$ and therefore fixes no line in $\mathbb{F}_q^2$; equivalently, t fixes no neighbor of v_0 in $\mathscr{B}$. Since $\mathscr{B}^t$ is a convex subtree containing v_0, we have $\mathscr{B}^t = \{v_0\}$. Because $\gcd(n, q+1) = 1$ and n is odd, $\rho_0(t^n) = \rho_0(t)^n \in \bar{T} \setminus \{\pm I\}$, so $\mathscr{B}^{t^n} = \{v_0\}$ as well.

(ii) Suppose $\rho_0(t) = zI$ and write $t = zI(I + \varpi^{r}A)$ as in Lemma 6.2. Any vertex $v \in \mathscr{B}$ at distance $d \geq 0$ from v_0 has the form $v = kD^{d}v_0$ for some $k \in K^{(0)}$. Write $A' = k^{-1}Ak = \left(\begin{smallmatrix} a'_{11} & a'_{12} \\ a'_{21} & a'_{22} \end{smallmatrix}\right) \in M_2(\mathcal{O}_F)$. Conjugating $zI \cdot t$ by kD^{d} gives

$$D^{-d}k^{-1}(zI \cdot t)kD^{d} = I + \varpi^{r}\begin{pmatrix} a'_{11} & \varpi^{-d}a'_{12} \\ \varpi^{d}a'_{21} & a'_{22} \end{pmatrix}. \tag{6.5}$$

By Lemma 6.2, $a'_{12} \in \mathcal{O}_F^{\times}$. The vertex $v = kD^{d}v_0$ is fixed by t if and only if the matrix (6.5) belongs to $\mathrm{SL}_2(\mathcal{O}_F)$, which holds if and only if $d \leq r$. This proves $\mathscr{B}^t = B(v_0, r)$.

To identify the reduction $\rho_j(x^{-1}tx)$ for $v = xv_j \in B(v_0, r)$ with $x \in G$ and $d = d(v, v_0) \equiv j \pmod 2$, write $d = 2m + j$ with $m \geq 0$. Since $\varpi^{-m}D^{2m} = \operatorname{diag}(\varpi^{m}, \varpi^{-m}) \in G$ acts on $\mathscr{B}$ and

by conjugation on G in the exact same way as D^{2m}, we may take $x = k\varpi^{-m}D^{2m} \in G$, so that $xv_j = kD^{2m}D^j v_0 = kD^d v_0 = v$. Then

$$D^{-j}x^{-1}txD^j = D^{-d}k^{-1}tkD^d = zI\left(I + \begin{pmatrix} \varpi^r a'_{11} & \varpi^{r-d}a'_{12} \\ \varpi^{r+d}a'_{21} & \varpi^r a'_{22} \end{pmatrix}\right).$$

Reducing modulo $\mathfrak{p}_F$ gives $\rho_j(x^{-1}tx) = \overline{D^{-j}x^{-1}txD^j}$. When $d < r$, all four entries of the perturbation matrix vanish modulo $\mathfrak{p}_F$, so $\rho_j(x^{-1}tx) = zI$. When $d = r$, the $(1,1)$, $(2,1)$, and $(2,2)$ entries vanish modulo $\mathfrak{p}_F$ while the $(1,2)$ entry reduces to $\overline{a'_{12}} \in \mathbb{F}_q^\times$, so $\rho_j(x^{-1}tx) = zU$ with $U = \left(\begin{smallmatrix} 1 & \overline{a'_{12}} \\ 0 & 1 \end{smallmatrix}\right)$ nontrivial unipotent.

Finally, expanding $t^n = z^n I(I + \varpi^r A)^n = zI(I + n\varpi^r A + O(\varpi^{2r}))$ and noting that $n\bar{A} \neq 0$ in $M_2(\mathbb{F}_q)$ since $\gcd(n, p) = 1$, we obtain $r_0(t^n) = r$ and $\mathscr{B}^{t^n} = B(v_0, r)$. □

7 Proofs of the character relations and closed formulas

The two comparison identities follow by matching the fixed-coset character sums for the ordinary and genuine compact inductions. The same fixed-ball counts then give the closed formulas and their two-orbit cancellation.

Let $\pi_{j,\theta} = \text{c-Ind}_{K^{(j)}}^G \sigma_\theta^{(j)}$ be the depth-zero supercuspidal representation induced from the vertex stabilizer $K^{(j)}$. By the Frobenius formula for the Harish-Chandra character of a compactly induced supercuspidal representation at a regular elliptic element $t \in G$ (see Sally [Sa88, Thm. 1.9 and Eq. (1.10)] and Bushnell–Henniart [BH06, Appendix A, §A.3, Thm. A.4]), the value $\Theta_{\pi_{j,\theta}}(t)$ is given by the finite sum over left cosets

$$\Theta_{\pi_{j,\theta}}(t) = \sum_{\substack{xK^{(j)} \in G/K^{(j)} \\ x^{-1}tx \in K^{(j)}}} \chi_\theta\big(\rho_j(x^{-1}tx)\big), \tag{7.1}$$

where an empty sum is zero. Under the G-equivariant bijection $xK^{(j)} \mapsto xv_j$ from $G/K^{(j)}$ onto the vertex orbit Gv_j, the condition $x^{-1}tx \in K^{(j)}$ is equivalent to $xv_j \in \mathscr{B}^t \cap Gv_j$, which is a finite set because t is regular elliptic. Note that $\Theta_{\pi_{j,\theta}}(t)$ as a pointwise function on G_{reg} is independent of the choice of Haar measure on G (which enters equally on both sides of the distribution pairing $\operatorname{tr}\pi(f) = \int_G f(g)\Theta_\pi(g)\,dg$).

The same trace calculation applies on the finite central cover $\widetilde{G}$. Recall that for an ε-genuine representation $\widetilde{\pi}$ of $\widetilde{G}$, the distribution character is paired against ε-anti-genuine locally constant compactly supported test functions $f(\zeta\widetilde{g}) = \varepsilon(\zeta)^{-1}f(\widetilde{g})$ ($\zeta \in \mu_n$), so that $\Theta_{\widetilde{\pi}}(\zeta\widetilde{g}) = \varepsilon(\zeta)\Theta_{\widetilde{\pi}}(\widetilde{g})$ on $\widetilde{G}_{\text{reg}}$. Since $\widetilde{G}/\widetilde{K}^{(j)} \cong G/K^{(j)}$, evaluating $\Theta_{\widetilde{\pi}_{j,\eta}}$ at $s_i(t)$ for $t \in T^{(i),\text{reg}}$ gives the finite sum

$$\Theta_{\widetilde{\pi}_{j,\eta}}\big(s_i(t)\big) = \sum_{\substack{xK^{(j)} \in G/K^{(j)} \\ x^{-1}tx \in K^{(j)}}} \operatorname{tr}\widetilde{\sigma}_{j,\eta}\big(\widetilde{x}^{-1}s_i(t)\widetilde{x}\big), \tag{7.2}$$

where $\widetilde{x} \in \widetilde{G}$ is any lift of x.

Proof of Theorem 1.2. Fix $i, j \in \{0,1\}$ and $t \in T^{(i),\mathrm{reg}}$. We first prove the direct character equality (1.12) for any regular character θ of $\bar{T}$.

Case 1: $\rho_i(t) \notin \{\pm I\}$. By Proposition 6.3(i), $\mathscr{B}^t = \{v_i\}$. If $j \neq i$, then $\mathscr{B}^t \cap Gv_j = \emptyset$, so both sums (7.1) and (7.2) are empty and both sides of (1.12) equal 0. If $j = i$, the unique contributing left coset is the identity coset $K^{(i)}$ (with $x = I$ and $\widetilde{x} = 1$), so

$$\Theta_{\widetilde{\pi}_{i,\theta}}\big(s_i(t)\big) = \operatorname{tr}\widetilde{\sigma}_{i,\theta}\big(s_i(t)\big) = \chi_\theta\big(\rho_i(t)\big) = \Theta_{\pi_{i,\theta}}(t).$$

Case 2: $\rho_i(t) = zI$ with $z \in \{\pm 1\}$. Here $t = zI \cdot h$ with $h \in K_+^{(i)}$. For every left coset $xK^{(j)} \in G/K^{(j)}$ such that $x^{-1}tx \in K^{(j)}$, Lemma 5.1(iii) yields

$$\widetilde{x}^{-1} s_i(t) \widetilde{x} = s_j(x^{-1}tx).$$

Applying $\operatorname{tr}\widetilde{\sigma}_{j,\theta}$ gives $\operatorname{tr}\widetilde{\sigma}_{j,\theta}(\widetilde{x}^{-1}s_i(t)\widetilde{x}) = \chi_\theta(\rho_j(x^{-1}tx))$. Thus the sums (7.1) and (7.2) have the exact same index set and identical summands term by term, proving (1.12).

To deduce the n-th power relation (1.13) for $\eta = \theta^{\langle n\rangle}$, apply (1.12) with η in place of θ and t^n in place of t to obtain

$$\Theta_{\widetilde{\pi}_{j,\eta}}\big(s_i(t^n)\big) = \Theta_{\pi_{j,\eta}}(t^n).$$

It therefore remains to check that $\Theta_{\pi_{j,\eta}}(t^n) = \Theta_{\pi_{j,\theta}}(t)$. When $\rho_i(t) = \bar{\tau} \notin \{\pm I\}$, Proposition 6.3(i) shows that both sides vanish for $j \neq i$, while for $j = i$ equations (1.9) and (5.3) give

$$\Theta_{\pi_{i,\eta}}(t^n) = \chi_\eta(\bar{\tau}^n) = -\eta(\bar{\tau}^n) - \eta(\bar{\tau}^{-n}) = -\theta(\bar{\tau}) - \theta(\bar{\tau}^{-1}) = \chi_\theta(\bar{\tau}) = \Theta_{\pi_{i,\theta}}(t).$$

When $\rho_i(t) = zI$, Proposition 6.3(ii) shows that $\mathscr{B}^{t^n} = \mathscr{B}^t = B(v_i, r_i(t))$, and that at each vertex $xv_j \in B(v_i, r_i(t)) \cap Gv_j$, both $\rho_j(x^{-1}tx)$ and $\rho_j(x^{-1}t^nx)$ equal zI if $d(xv_j, v_i) < r_i(t)$ and equal z times a nontrivial unipotent matrix if $d(xv_j, v_i) = r_i(t)$. By Proposition 5.2(ii)–(iii) and $\eta(zI) = \theta(zI)$ from (1.10), each fixed vertex contributes the same value to $\Theta_{\pi_{j,\eta}}(t^n)$ and to $\Theta_{\pi_{j,\theta}}(t)$. This completes the proof of Theorem 1.2. □

Proof of Theorem 1.3. Part (i) was established in Case 1 of the proof of Theorem 1.2 via Proposition 5.2(i).

For part (ii), suppose $\rho_i(t) = zI$ with $z \in \{\pm 1\}$ and $r = r_i(t) \geq 1$. By Proposition 6.3(ii), the contributing vertices in (7.1) are those $v \in B(v_i, r)$ with $d = d(v, v_i) \equiv j - i \pmod 2$. By Proposition 5.2(ii)–(iii) and Proposition 6.3(ii), each vertex at distance $d < r$ contributes $(q-1)\theta(zI)$, while each boundary vertex at distance $d = r$ contributes $-\theta(zI)$.

Subcase 2a: $j = i$ (even distances $d = 2k \leq r$). If $r = 2m + 1$ is odd, every even distance $d = 2k \leq r$ satisfies $d < r$, so (6.1) gives

$$\Theta_{\pi_{i,\theta}}(t) = \theta(zI)(q-1)\left[1 + \sum_{k=1}^{m}(q+1)q^{2k-1}\right] = \theta(zI)\big(q^{2m+1} - 1\big).$$

If $r = 2m$ is even, the sphere $d = 2m$ lies on the boundary of $B(v_i, r)$, so

$$\Theta_{\pi_{i,\theta}}(t) = \theta(zI)\left[(q-1)\left(1 + \sum_{k=1}^{m-1}(q+1)q^{2k-1}\right) - (q+1)q^{2m-1}\right] = \theta(zI)\big(-q^{2m} - 1\big).$$

Subcase 2b: $j \neq i$ (odd distances $d = 2k+1 \leq r$). If $r = 2m$ is even, every odd distance $d = 2k+1 \leq r$ satisfies $d < r$, so

$$\Theta_{\pi_{j,\theta}}(t) = \theta(zI)(q-1)\sum_{k=0}^{m-1}(q+1)q^{2k} = \theta(zI)\big(q^{2m}-1\big).$$

If $r = 2m+1$ is odd, the sphere $d = 2m+1$ is the boundary of $B(v_i, r)$, yielding

$$\Theta_{\pi_{j,\theta}}(t) = \theta(zI)\left[(q-1)\sum_{k=0}^{m-1}(q+1)q^{2k} - (q+1)q^{2m}\right] = \theta(zI)\big(-q^{2m+1}-1\big).$$

All four cases combine into the single expression $\theta(zI)\big((-1)^{r+1+j-i}q^r - 1\big)$ of (1.15).

Finally, part (iii) follows immediately by summing (1.14) or (1.15) over $j \in \{0,1\}$ and applying (1.13). □

Example. In the $\mathbb{Q}_7$ example with central reduction I at depth $r = 1$, the even orbit contributes 6 and the odd orbit contributes -8. Thus their sum is -2, as in (1.16). Section 8 provides the fixed-vertex computation.

Remark 7.1 (Interpretation via restriction from $\mathrm{GL}_2(F)$ and depth-zero L-packets). The cancellation of the $\pm q^r$ terms in (1.16) has a conceptual explanation via restriction from $\mathrm{GL}_2(F)$. Choose a character of $\mathbb{F}_{q^2}^\times$ extending θ; the associated cuspidal representation Σ_θ of $\mathrm{GL}_2(\mathbb{F}_q)$ restricts irreducibly to σ_θ on $\mathrm{SL}_2(\mathbb{F}_q)$ because $\theta^2 \neq 1$. Inflate Σ_θ to $\mathrm{GL}_2(\mathcal{O}_F)$, extend it to $F^\times\mathrm{GL}_2(\mathcal{O}_F)$ by letting $F^\times$ act via a character compatible with the central character of Σ_θ, and put $\Pi_\theta = \text{c-Ind}_{F^\times\mathrm{GL}_2(\mathcal{O}_F)}^{\mathrm{GL}_2(F)}\Sigma_\theta$. Then Π_θ is an irreducible depth-zero supercuspidal representation of $\mathrm{GL}_2(F)$. Because the double coset space $G\backslash\mathrm{GL}_2(F)/F^\times\mathrm{GL}_2(\mathcal{O}_F)$ has exactly two elements, with representatives I and D, Mackey's restriction formula gives

$$\Pi_\theta|_G \cong \pi_{0,\theta} \oplus \pi_{1,\theta},$$

which is the standard depth-zero supercuspidal L-packet of $G = \mathrm{SL}_2(F)$ associated with θ. Moreover, the coset space $\mathrm{GL}_2(F)/F^\times\mathrm{GL}_2(\mathcal{O}_F)$ is in bijection with the entire vertex set $\mathscr{B} = Gv_0 \sqcup Gv_1$. Computing $\Theta_{\Pi_\theta}(t) = \Theta_{\pi_{0,\theta}}(t) + \Theta_{\pi_{1,\theta}}(t)$ for $t \in T^{(i),\mathrm{reg}}$ with $\rho_i(t) = zI$ and $r = r_i(t) \geq 1$ therefore sums directly over all vertices of $B(v_i, r)$ without parity restriction:

$$\Theta_{\Pi_\theta}(t) = \theta(zI)\left[(q-1)\left(1 + \sum_{d=1}^{r-1}(q+1)q^{d-1}\right) - (q+1)q^{r-1}\right] = -2\theta(zI),$$

since the interior vertices contribute $(q^r + q^{r-1} - 2)\theta(zI)$ and the boundary sphere contributes $-(q^r + q^{r-1})\theta(zI)$. In the notation of Sally–Shalika [SS68] and Aubert–Plymen [AP24], the individual constituent characters near I take the form $-1 \pm f(t)$ with $f(t) = (-1)^r q^r$, whose stable packet sum is -2.

Remark 7.2 (Weyl discriminant and root-sum formulation). For $t \in T^{(i),\mathrm{reg}}$, let $\Delta(t) = \big|\det(1-\mathrm{Ad}(t) \mid \mathfrak{sl}_2(F)/\mathrm{Lie}(T^{(i)}))\big|_F^{1/2}$ be the Weyl discriminant. When $\rho_i(t) \notin \{\pm I\}$, $\Delta(t) = \Delta(t^n) = 1$; when $\rho_i(t) = zI$, the two nontrivial root eigenvalues of $\mathrm{Ad}(t)$ have distance from 1 of valuation $r_i(t)$, so

$\Delta(t) = \Delta(t^n) = q^{-r_i(t)}$. Multiplying (1.13) by $\Delta(t^n)$ and using the uniqueness of n-th roots in G for $t^n \in T^{(i),\mathrm{reg}}$ (Lemma 6.1) allows one to write (1.13) in the formal Flicker–Kazhdan root-sum shape

$$\Delta(t^n)\Theta_{\widetilde{\pi}_{j,\theta\langle n\rangle}}\big(s_i(t^n)\big) = \sum_{\substack{h\in G\\ h^n=t^n}} \Delta(h)\Theta_{\pi_{j,\theta}}(h),$$

where the sum on the right-hand side has the single term $h = t$.

8 Worked example over $\mathbb{Q}_7$

This running example makes the two vertex orbits, the power-map reparameterization, and the fixed-ball formulas explicit. It also provides numerical checks of the values appearing in the preceding sections.

Take $F = \mathbb{Q}_7$ ($p = q = 7$) and covering degree $n = 3$, so that $n \mid (q-1) = 6$ and $\gcd(n, q+1) = \gcd(3, 8) = 1$. Since 3 is a quadratic nonresidue modulo 7, the quadratic extension $E = \mathbb{Q}_7(\sqrt{3})$ is unramified. Using the $\mathbb{Z}_7$-basis $(1, \sqrt{3})$ of $O_E = \mathbb{Z}_7[\sqrt{3}]$, multiplication by $\alpha + \beta\sqrt{3} \in E$ embeds $T^{(0)}$ into $K^{(0)} = \mathrm{SL}_2(\mathbb{Z}_7)$ as

$$\alpha + \beta\sqrt{3} \longmapsto \begin{pmatrix} \alpha & 3\beta \\ \beta & \alpha \end{pmatrix} \qquad (\alpha^2 - 3\beta^2 = 1).$$

The finite nonsplit torus $\bar{T} \subset \mathrm{SL}_2(\mathbb{F}_7)$ is cyclic of order $q+1 = 8$. Fix a generator $\gamma_0 \in \bar{T}$ and define $\theta : \bar{T} \to \mathbb{C}^\times$ by $\theta(\gamma_0) = e^{2\pi i/8}$. Since $3^{-1} \equiv 3 \pmod 8$, the reparameterized character in (1.9) is $\eta = \theta^{\langle 3\rangle} = \theta^3$.

For any lift $t_0 \in T^{(0)}$ of γ_0, both t_0 and t_0^3 have regular elliptic reduction in $\bar{T}\setminus\{\pm I\}$. Theorem 1.3(i) and Theorem 1.2 give

$$\Theta_{\widetilde{\pi}_{0,\eta}}\big(s_0(t_0^3)\big) = \Theta_{\pi_{0,\theta}}(t_0) = -2\cos(\pi/4) = -\sqrt{2}, \qquad \Theta_{\widetilde{\pi}_{1,\eta}}\big(s_0(t_0^3)\big) = \Theta_{\pi_{1,\theta}}(t_0) = 0.$$

To illustrate the central-reduction regime across depths $r \geq 1$, consider the elements

$$t_r = \frac{1 + 7^r\sqrt{3}}{1 - 7^r\sqrt{3}} \in T^{(0),\mathrm{reg}} \qquad (r \geq 1). \tag{8.1}$$

Expanding $(1 + 7^r\sqrt{3})(1 - 7^r\sqrt{3})^{-1} = 1 + 2\cdot 7^r\sqrt{3} + O(7^{2r})$ shows that $t_r - I$ has leading term $2\cdot 7^r \left(\begin{smallmatrix} 0 & 3 \\ 1 & 0 \end{smallmatrix}\right)$, so $\rho_0(t_r) = I$, $\theta(I) = 1$, and $r_0(t_r) = r_0(t_r^3) = r$. In the 7-adic Bruhat–Tits tree (which is 8-regular), the spheres around v_0 have sizes $N_0 = 1$, $N_1 = 8$, $N_2 = 56$, and $N_3 = 392$. Table 1 summarizes the fixed-vertex counts and the resulting ordinary and genuine character values for $r = 1, 2, 3$.

Each entry reflects the interior versus boundary contributions in Theorem 1.3(ii):

- For $r = 1$, the base vertex v_0 is strictly interior ($d = 0 < 1$) and contributes $\chi_\theta(I) = 6$ to the even orbit $j = 0$, whereas the 8 neighbors at distance $d = 1$ lie on the boundary and contribute -1 each to the odd orbit $j = 1$, giving -8.
- For $r = 2$, the 8 vertices at distance $d = 1$ become strictly interior and contribute $8 \times 6 = 48$ to the odd orbit $j = 1$, while the 56 vertices at distance $d = 2$ form the boundary and contribute -56 to the even orbit $j = 0$, giving $6 - 56 = -50$.

Table 1: Fixed-ball vertex counts and character values at $t_r \in T^{(0),\mathrm{reg}}$ over $\mathbb{Q}_7$ for $n = 3$. By Theorem 1.2, $\Theta_{\pi_{j,\theta}}(t_r) = \Theta_{\widetilde{\pi}_{j,\eta}}(s_0(t_r^3))$ for $j = 0, 1$.

r	$\lvert B(v_0, r) \cap Gv_0 \rvert$	$\lvert B(v_0, r) \cap Gv_1 \rvert$	$\Theta_{\pi_{0,\theta}}(t_r)$	$\Theta_{\pi_{1,\theta}}(t_r)$	Packet sum
1	1	8	6	−8	−2
2	1 + 56 = 57	8	−50	48	−2
3	1 + 56 = 57	8 + 392 = 400	342	−344	−2

- For $r = 3$, the 57 even-distance vertices at $d \in \{0, 2\}$ are all strictly interior and contribute $57 \times 6 = 342$ to $j = 0$, while for $j = 1$ the 8 interior vertices at $d = 1$ contribute 48 and the 392 boundary vertices at $d = 3$ contribute -392, giving $48 - 392 = -344$.

9 Open problems and next steps

Several natural problems arise when moving beyond the unramified elliptic generic depth-zero setting treated here:

(i) *The split maximal torus and principal series comparison.* As noted in Remark 1.4, the preimage of the unramified torus $T^{(i)}$ in $\widetilde{G}$ is abelian, so every element of $T^{(i)}$ is good and the root sum over $\{h : h^n = t^n\}$ has a single term. On the split diagonal torus $A(F) \subset \mathrm{SL}_2(F)$, by contrast, the metaplectic commutator pairing is nontrivial, genuine characters are supported on n-th powers modulo the center, and the equation $h^n = a^n$ has n solutions in $A(F)$ (for a cover whose commutator pairing on $A(F)$ is nondegenerate modulo n-th powers). Verifying the n-term Flicker–Kazhdan relation on the compact regular elements of $A(F)$, where supercuspidal characters need not vanish, is the first genuine test of the n-th power support mechanism.

(ii) *Exceptional depth-zero representations (quadratic θ).* When $\theta : \bar{T} \to \{\pm 1\}$ is the unique nontrivial quadratic character of $\bar{T}$, the Deligne–Lusztig representation $-R_{\bar{T}}^{\mathrm{SL}_2(\mathbb{F}_q)}(\theta)$ splits into two irreducible cuspidal representations σ_θ^+ and σ_θ^- of degree $(q-1)/2$ whose values on the two rational unipotent conjugacy classes of $\mathrm{SL}_2(\mathbb{F}_q)$ differ by a quadratic Gauss sum. However, their sum $\sigma_\theta^+ \oplus \sigma_\theta^-$ takes the constant value -1 on both unipotent classes, so the packet-sum comparison extends naturally even though the individual constituent comparison requires tracking how the n-th power map interacts with rational unipotent classes.

(iii) *Ramified elliptic tori and positive-depth supercuspidal representations.* Ramified elliptic maximal tori lie in Iwahori subgroups (they fix an edge of $\mathscr{B}$ and reduce into $\{\pm I\}$ times a unipotent group at each of its vertices), so the fixed-ball analysis must be recentered at the midpoint of an edge, and positive-depth supercuspidal representations involve characters of Moy–Prasad filtration quotients where Gauss sums and Weil indices enter the character formulas [SS68, DS18].

Example of a case outside the theorem. The $\mathbb{Q}_7$ calculation of Section 8 concerns an unramified elliptic torus. Replacing it by a ramified elliptic torus would require a separate analysis of its fixed edge, as described in the preceding open-problems list.

10 Summary of results

We summarize the results proved in this paper, distinguishing the established unramified elliptic case from the directions not treated. No new character identity is asserted here.

(i) Proposition 1.1 establishes unique compact vertex splittings for the tame odd-degree covers under the stated hypotheses.

(ii) Theorem 1.2 identifies the ordinary and genuine characters on both unramified elliptic torus classes and gives the reparameterized n-th power identity.

(iii) Theorem 1.3 supplies explicit fixed-ball formulas. Its sum over the two vertex orbits is independent of the depth when reduction is central.

Example. For $F = \mathbb{Q}_7$ and $n = 3$, the two characters at depth $r = 1$ take the values 6 and -8, whose sum is -2. The same cancellation appears for $r = 2, 3$ in Table 1.

Glossary of notation and terminology

This reference list recalls specialist terms used in the body. Sections 3 and 4 provide context and the recurring $\mathbb{Q}_7$ examples.

Key terminology

The following entries give concise meanings of the technical terms introduced in the main text.

Anti-genuine test function
A locally constant, compactly supported function $f : \widetilde{G} \to \mathbb{C}$ satisfying $f(\zeta\widetilde{g}) = \varepsilon(\zeta)^{-1} f(\widetilde{g})$ for all $\zeta \in \mu_n$ and $\widetilde{g} \in \widetilde{G}$, used so that the integrand $f(\widetilde{g})\Theta_{\widetilde{\pi}}(\widetilde{g})$ descends to $G = \widetilde{G}/\mu_n$.

Bruhat–Tits tree ($\mathscr{B}$)
The $(q + 1)$-regular simplicial tree whose vertices are homothety classes $[L]$ of rank-two O_F-lattices $L \subset F^2$, with an edge between $[L]$ and $[L']$ whenever $\varpi L \subsetneq L' \subsetneq L$.

Deligne–Lusztig cuspidal representation (σ_θ)
An irreducible $(q - 1)$-dimensional representation of $\mathrm{SL}_2(\mathbb{F}_q)$ having no nonzero fixed vectors under any nontrivial unipotent subgroup, parameterized by a regular character $\theta : \bar{T} \to \mathbb{C}^\times$ ($\theta^2 \neq 1$).

Depth-zero supercuspidal representation
An irreducible supercuspidal representation of G (or $\widetilde{G}$) possessing nonzero vectors fixed by the pro-p first congruence subgroup $K_+^{(j)}$ (or its canonical lift $s_j(K_+^{(j)})$) for a vertex v_j of $\mathscr{B}$.

Genuine (ε-genuine) representation
A representation $(\widetilde{\pi}, V)$ of the central cover $\widetilde{G}$ on which the central kernel μ_n acts via a fixed injective character $\varepsilon : \mu_n \hookrightarrow \mathbb{C}^\times$, i.e., $\widetilde{\pi}(\zeta\widetilde{g}) = \varepsilon(\zeta)\widetilde{\pi}(\widetilde{g})$.

Good element of a covering group
An element $g \in G$ whose centralizer $Z_G(g)$ has abelian preimage $\mathbf{p}^{-1}(Z_G(g))$ in $\widetilde{G}$; genuine characters can be nonzero only on lifts of good elements.

Harish-Chandra character (Θ_π)
The locally integrable, conjugation-invariant function on the regular semisimple set G_{reg} (smooth and locally constant on G_{reg}) representing the trace distribution $\operatorname{tr}\pi(f) = \int_G f(g)\Theta_\pi(g)\,dg$.

L-packet (depth-zero)
A finite set of irreducible representations of $G = \mathrm{SL}_2(F)$ constituting a single restriction packet from $\mathrm{GL}_2(F)$ (here $\{\pi_{0,\theta}, \pi_{1,\theta}\}$ for $\theta^2 \neq 1$) whose sum of characters is stable on regular elliptic elements.

Tame central cover
A central extension $1 \to \mu_n \to \widetilde{G} \to G \to 1$ whose kernel order n is coprime to the residue characteristic p of F.

Acknowledgments and provenance

This article reconstructs ideas originating in David Joyner's earlier unfinished manuscript *Character relations for supercuspidal representations of* $\overline{SL(2)}$ (surviving in the source files `CHART.TEX`, `CHAR0.TEX`, `CHAR1.TEX`, and `CHAR2.TEX`) and its published companion [Jo99]. ChatGPT (Sol, OpenAI) assisted with earlier drafts, with the unified exposition, and with the present editorial style revision. Argon (Gemini, Google) prepared the mathematical version underlying this style revision and provided an earlier referee report. Fable (Claude, Anthropic) supplied a referee report identifying the cohomological existence and uniqueness of compact vertex splittings, the structural role of the abelian unramified torus preimage, and the $\mathrm{GL}_2(F)$ restriction interpretation of the two-orbit sum. AI review is not an independent certification of the mathematical results. Further details of the AI usage, earlier draft history, and changes made in this style revision are recorded in `revision_log.md`.

References

[Ad03] J. Adams, *Characters of covering groups of* $\mathrm{SL}(n)$, J. Inst. Math. Jussieu **2** (2003), no. 1, 1–21 (preprint 1998).

[AP24] A.-M. Aubert and R. Plymen, *Comparison of the Sally–Shalika character formulas with the endoscopic character identities for* SL_2, arXiv:2410.20183 [math.RT] (2024; rev. 2025).

[Bl85] C. Blondel, *Les représentations supercuspidales des groupes métaplectiques sur* $\mathrm{GL}(2)$ *et leurs caractères*, Mém. Soc. Math. France (N.S.) **18** (1985), 1–119.

[Bo11] C. Bonnafé, *Representations of* $\mathrm{SL}_2(\mathbb{F}_q)$, Algebra and Applications, vol. 13, Springer-Verlag, London, 2011.

[BD01] J.-L. Brylinski and P. Deligne, *Central extensions of reductive groups by* $\mathbf{K}_2$, Publ. Math. Inst. Hautes Études Sci. **94** (2001), 5–85.

[BH06] C. J. Bushnell and G. Henniart, *The Local Langlands Conjecture for* $\mathrm{GL}(2)$, Grundlehren der mathematischen Wissenschaften, vol. 335, Springer-Verlag, Berlin, 2006.

[Ca14] W. Casselman, *The Bruhat–Tits tree of* SL_2, lecture notes, University of British Columbia (2014).

[CG09] C. Cunningham and J. Gordon, *Motivic proof of a character formula for* $\mathrm{SL}(2)$, Experiment. Math. **18** (2009), no. 1, 11–44.

[DS18] S. DeBacker and L. Spice, *Stability of character sums for positive-depth, supercuspidal representations*, J. Reine Angew. Math. **742** (2018), 47–78.

[DL76] P. Deligne and G. Lusztig, *Representations of reductive groups over finite fields*, Ann. of Math. (2) **103** (1976), 103–161.

[DM20] F. Digne and J. Michel, *Representations of Finite Groups of Lie Type*, 2nd ed., London Mathematical Society Student Texts, vol. 95, Cambridge University Press, Cambridge, 2020.

[Fl80] Y. Z. Flicker, *Automorphic forms on covering groups of* $\mathrm{GL}(2)$, Invent. Math. **57** (1980), no. 2, 119–182.

[FK86] Y. Z. Flicker and D. A. Kazhdan, *Metaplectic correspondence*, Publ. Math. Inst. Hautes Études Sci. **64** (1986), 53–110.

[HW09] T. K. Howard and M. H. Weissman, *Depth-zero representations of nonlinear covers of p-adic groups*, Int. Math. Res. Not. IMRN **2009**, no. 21, 3979–3995.

[Jo99] D. Joyner, *A correspondence for the supercuspidal representations of the metaplectic cover of* $\mathrm{SL}(2, F)$*,* F *p-adic*, Arch. Math. (Basel) **73** (1999), no. 5, 332–340.

[KP84] D. A. Kazhdan and S. J. Patterson, *Metaplectic forms*, Publ. Math. Inst. Hautes Études Sci. **59** (1984), 35–142.

[Ku67] T. Kubota, *Topological covering of* $\mathrm{SL}(2)$ *over a local field*, J. Math. Soc. Japan **19** (1967), 114–121.

[Li12] W.-W. Li, *La formule des traces pour les revêtements de groupes réductifs connexes. II. Analyse harmonique locale*, Ann. Sci. Éc. Norm. Supér. (4) **45** (2012), no. 5, 787–859.

[Ma69] H. Matsumoto, *Sur les sous-groupes arithmétiques des groupes semi-simples déployés*, Ann. Sci. Éc. Norm. Supér. (4) **2** (1969), 1–62.

[Mo68] C. C. Moore, *Group extensions of p-adic and adelic linear groups*, Publ. Math. Inst. Hautes Études Sci. **35** (1968), 5–70.

[MP94] A. Moy and G. Prasad, *Unrefined minimal K-types for p-adic groups*, Invent. Math. **116** (1994), no. 1–3, 393–408.

[MP96] A. Moy and G. Prasad, *Jacquet functors and unrefined minimal K-types*, Comment. Math. Helv. **71** (1996), no. 1, 98–121.

[Sa88] P. J. Sally, Jr., *Some remarks on discrete series characters for reductive p-adic groups*, in *Representations of Lie Groups, Kyoto, Hiroshima, 1986*, Advanced Studies in Pure Mathematics, vol. 14, Kinokuniya, Tokyo, 1988, pp. 337–348.

[SS68] P. J. Sally, Jr. and J. A. Shalika, *Characters of the discrete series of representations of* SL(2) *over a local field*, Proc. Nat. Acad. Sci. U.S.A. **61** (1968), 1231–1237.

[Wa80] J.-L. Waldspurger, *Correspondance de Shimura*, J. Math. Pures Appl. (9) **59** (1980), no. 1, 1–132.

[Wa91] J.-L. Waldspurger, *Correspondances de Shimura et quaternions*, Forum Math. **3** (1991), no. 3, 219–307.

[We09] M. H. Weissman, *Metaplectic tori over local fields*, Pacific J. Math. **241** (2009), no. 1, 169–200.